Lecture 0

Prerequisites: A Refresher

This course assumes you can already *compute* with the tools of a first linear-algebra or methods course; from Lecture 1 onward we build abstraction on top of that fluency rather than re-teaching it. The four topics below are the ones the early lectures lean on hardest. Complex arithmetic is used on the very first pages of Lecture 1 and becomes load-bearing from Week 7 (inner products, Pauli matrices, quantum phases); matrix algebra, determinants, and Gaussian elimination underpin everything from bases and rank to the characteristic polynomial of Week 5. None of this is new material—it is a fast recall pass, paired with the companion self-check (`exercises-0.pdf`) so you can find any rust before it costs you. Work the self-check first; use the sections here to repair whatever it exposes.

Learning objectives

By the end of this lecture you will be able to:

- Add, multiply, conjugate, and divide complex numbers, and move freely between Cartesian and polar form.
- Multiply matrices, transpose them, and read off the trace, knowing which operations commute and which do not.
- Evaluate 2×2 and 3×3 determinants and use $\det A \neq 0$ as the test for invertibility.
- Row-reduce a linear system, describe its solution set, and find a basis for the solutions of $Ax = 0$.

1 Complex numbers

Quantum mechanics lives over $\mathbb{C}$, so complex arithmetic is not optional background—it is the air the second half of the course breathes. A *complex number* is written $z = a + bi$ with $a, b \in \mathbb{R}$ and $i^2 = -1$; its *real* and *imaginary* parts are $\Re z = a$ and $\Im z = b$. Addition is componentwise and multiplication follows from $i^2 = -1$ and the distributive law, e.g. $(2+i)(1+3i) = 2 + 6i + i + 3i^2 = -1 + 7i$.

1.1 Conjugate, modulus, division

The *conjugate* of $z = a + bi$ is $\overline{z} = a - bi$, and its *modulus* is

$$|z| = \sqrt{z\overline{z}} = \sqrt{a^2 + b^2}, \qquad \text{so} \qquad z\overline{z} = |z|^2.$$

Conjugation respects the arithmetic, $\overline{z+w} = \overline{z} + \overline{w}$ and $\overline{zw} = \overline{z}\,\overline{w}$, the modulus is multiplicative, $|zw| = |z|\,|w|$, and z is real exactly when $\overline{z} = z$. To *divide*, multiply top and bottom by the conjugate of the denominator, turning it real:

$$\frac{3+2i}{1-i} = \frac{(3+2i)(1+i)}{(1-i)(1+i)} = \frac{3+3i+2i+2i^2}{1^2+1^2} = \frac{1+5i}{2} = \tfrac{1}{2} + \tfrac{5}{2}i.$$

1.2 Polar form and Euler's formula

Every complex number has a *polar form*

$$z = r\,e^{i\theta} = r(\cos\theta + i\sin\theta), \qquad r = |z|, \quad \theta = \arg z, \tag{0.1}$$

where (0.1) is *Euler's formula*. In polar form multiplication is almost free—moduli multiply and arguments add,

$$r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2\, e^{i(\theta_1+\theta_2)}, \qquad \text{so} \qquad \left(re^{i\theta}\right)^n = r^n e^{in\theta}$$

(the second identity is *De Moivre's theorem*).

Example 0.1 (Powers via polar form). Write $-1+\sqrt{3}\,i$ in polar form and cube it. Here $r = \sqrt{1+3} = 2$ and the point lies in the second quadrant at $\theta = \frac{2\pi}{3}$, so $-1+\sqrt{3}\,i = 2\,e^{i2\pi/3}$. Then

$$\left(-1+\sqrt{3}\,i\right)^3 = 2^3 e^{i\cdot 2\pi} = 8.$$

Cubing tripled the argument to a full turn and cubed the modulus—one line in place of a binomial expansion. ◁

Remark 0.2. The factor $e^{i\theta}$ is a complex number of modulus 1: a pure *phase*. Phases are the mathematical home of quantum interference, and the n-th *roots of unity*—the solutions of $z^n = 1$, namely $e^{2\pi ik/n}$ for $k = 0, \dots, n-1$—are the phases that recur whenever a finite cyclic symmetry appears. This is why a real vector space cannot carry quantum theory; we will need $\mathbb{C}$ from Week 7 on.

2 Matrix algebra

A matrix in $M_{m\times n}(\mathbb{F})$ has m rows and n columns; matrices of the same shape add entrywise and scale entrywise. Two operations are worth recalling with care, because the course uses them constantly and because one of them does *not* behave like ordinary multiplication.

2.1 Multiplication

The product AB is defined only when the number of columns of A equals the number of rows of B; if A is $m\times n$ and B is $n\times p$ then AB is $m\times p$ with entries

$$(AB)_{ik} = \sum_{j=1}^{n} A_{ij} B_{jk}.$$

Multiplication is associative and distributes over addition, and the identity matrix I_n (ones on the diagonal, zeros elsewhere) satisfies $AI = IA = A$. But it is *not commutative*:

$$\begin{pmatrix}1&1\\0&1\end{pmatrix}\begin{pmatrix}1&0\\1&1\end{pmatrix} = \begin{pmatrix}2&1\\1&1\end{pmatrix}, \qquad \begin{pmatrix}1&0\\1&1\end{pmatrix}\begin{pmatrix}1&1\\0&1\end{pmatrix} = \begin{pmatrix}1&1\\1&2\end{pmatrix},$$

so in general $AB \neq BA$. This single fact—that operators need not commute—is what makes quantum mechanics quantum, so it is worth feeling in your fingers now.

2.2 Transpose and trace

The *transpose* A^T swaps rows and columns, $(A^\mathsf{T})_{ij} = A_{ji}$, and reverses products: $(AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}$. A matrix is *symmetric* if $A^\mathsf{T} = A$ and *antisymmetric* if $A^\mathsf{T} = -A$—two subspaces you will meet again in Lecture 1. The *trace* is the sum of the diagonal entries, $\operatorname{tr} A = \sum_i A_{ii}$; it is linear and satisfies the cyclic identity

$$\operatorname{tr}(AB) = \operatorname{tr}(BA), \tag{0.2}$$

even though $AB \neq BA$. The *inverse* A^{-1}, when it exists, is the square matrix with $AA^{-1} = A^{-1}A = I$, and inverses also reverse products, $(AB)^{-1} = B^{-1}A^{-1}$. *When* an inverse exists, and how to compute it, are the subjects of the next two sections.

Remark 0.3. Over $\mathbb{C}$ the operation that really matters is the *conjugate transpose* (or Hermitian adjoint) $A^\dagger = \overline{A}^\mathsf{T}$, which transposes and conjugates every entry. It is the matrix face of the adjoint $T^\dagger$ that organizes all of Weeks 9–10; for now simply note that the complex analogue of "symmetric" is $A^\dagger = A$ (Hermitian), not $A^\mathsf{T} = A$.

3 Determinants

To every *square* matrix the determinant assigns a scalar that detects invertibility. For 2×2 and 3×3 matrices the formulas are

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc, \qquad \det\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a\begin{vmatrix} e & f \\ h & i \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + c\begin{vmatrix} d & e \\ g & h \end{vmatrix},$$

the second being *cofactor (Laplace) expansion* along the top row. One may expand along any row or column, attaching the sign $(-1)^{i+j}$ to the entry in position (i, j); a good choice of row or column (one with zeros) saves work.

Example 0.4 (A 3×3 determinant). Expanding along the first row,

$$\det\begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix} = 2(3 \cdot 2 - 1 \cdot 1) - 1(1 \cdot 2 - 1 \cdot 0) + 0 = 2 \cdot 5 - 2 = 8 \neq 0,$$

so this matrix is invertible. ◁

3.1 The facts you will actually use

Three properties carry most of the weight. The determinant is *multiplicative*, $\det(AB) = \det A \det B$; it is unchanged by transpose, $\det A^\mathsf{T} = \det A$; and for a triangular matrix it is just the product of the diagonal entries. Row operations act predictably: swapping two rows flips the sign, scaling a row by c scales the determinant by c, and adding a multiple of one row to another leaves it unchanged (the basis of the next section). Above all,

$$\begin{aligned} A \text{ is invertible} &\iff \det A \neq 0 \\ &\iff \text{the columns of } A \text{ are linearly independent} \\ &\iff Ax = 0 \text{ only for } x = 0. \end{aligned}$$

For a 2×2 matrix the inverse is immediate from the determinant,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

while in larger sizes row reduction (Section 4) is the practical route.

Remark 0.5 (Looking ahead to Week 5). The determinant is how eigenvalues will enter. The *characteristic polynomial* of a square matrix A is $\det(\lambda I - A)$—a monic polynomial in λ whose roots, the values of λ making $A - \lambda I$ singular, are exactly the eigenvalues. (Week 5 uses this monic form throughout.) Because you already know determinants, we can introduce eigenvalues this way in Week 5 without delay, rather than building up to them as Axler does.

4 Gaussian elimination

Gaussian elimination is the workhorse for solving linear systems, and from Week 2 it is also how we compute bases, rank, and dimension. The method applies three *elementary row operations*—swap two rows, scale a row by a nonzero constant, add a multiple of one row to another—to drive a system toward *reduced row-echelon form* (RREF): each leading nonzero entry is a 1, sits to the right of the one above it, and is the only nonzero entry in its column.

4.1 Solving a system

Write the system as an augmented matrix $[A \mid b]$ and reduce. The shape of the result tells you everything: a pivot in every column of A gives a unique solution; a column without a pivot marks a *free variable* and hence infinitely many solutions; and a row $[\,0\ \cdots\ 0 \mid 1\,]$ signals an inconsistent system with no solution at all.

Example 0.6 (A unique solution). For

$$\begin{aligned} x + 2y + \ \ z &= 4, \\ 2x - \ \ y + 3z &= 4, \\ -x + \ \ y + 2z &= 2, \end{aligned} \qquad [A \mid b] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & -1 & 3 & 4 \\ -1 & 1 & 2 & 2 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right],$$

so $(x, y, z) = (1, 1, 1)$ is the unique solution. ◁

4.2 Homogeneous systems and the null space

The homogeneous system $Ax = 0$ is always consistent—$x = 0$ works—and it has *nontrivial* solutions precisely when there is a free variable. Its solution set is the *null space* of A, and it is a vector space (you will recognize it as one of the first examples in Lecture 1). Reading a basis off the RREF is a skill the course uses repeatedly.

Example 0.7 (A basis for the solution space). Take $A = \left(\begin{smallmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{smallmatrix}\right)$. The second row is twice the first, so the RREF is the single equation $x_1 + 2x_2 - x_3 = 0$. Treating $x_2 = s$ and $x_3 = t$ as free gives $x_1 = -2s + t$, hence

$$x = s\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

so $\{(-2, 1, 0),\ (1, 0, 1)\}$ is a basis for the null space, which is therefore two-dimensional. Here A has rank 1 and nullity 2, and $1 + 2 = 3$ is the number of columns—a first glimpse of the rank–nullity theorem of Week 3. ◁

Remark 0.8 (Inverses by row reduction). Row reduction also inverts a matrix: form the augmented block $[A \mid I]$ and reduce. If A is invertible the left half becomes I and the right half is then A^{-1}; if a zero row appears on the left, A is singular. This is the practical companion to the determinant test $\det A \neq 0$ of Section 3.

Summary of main concepts

The four skills above are the computational floor the course is built on. From $\mathbb{C}$ you need conjugate, modulus, division, and the polar form $re^{i\theta}$, because everything quantum is complex. From matrix algebra you need multiplication (non-commutative!), transpose, and trace. Determinants give the one-number test $\det A \neq 0$ for invertibility and, in Week 5, the characteristic polynomial $\det(\lambda I - A)$. Gaussian elimination solves systems and—reading a basis off the RREF of $Ax = 0$—computes the null spaces, ranks, and dimensions that Lectures 1–4 turn into abstract structure. If the self-check exposes a gap, repair it here before Lecture 1.

- **Complex arithmetic**—$\bar{z}$, $|z| = \sqrt{z\bar{z}}$, division by the conjugate, and $z = re^{i\theta}$ with moduli multiplying and arguments adding.
- **Matrix algebra**—$(AB)_{ik} = \sum_j A_{ij}B_{jk}$ (and $AB \neq BA$); transpose A^{T}; trace $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.
- **Determinants**—2×2 and 3×3 by cofactors; $\det(AB) = \det A \det B$; $\det A \neq 0 \iff$ invertible $\iff$ columns independent.

- **Gaussian elimination**—RREF, pivots vs. free variables; the null space of $Ax = 0$ and a basis for it; rank as the pivot count.

The companion self-check is in `exercises-0.pdf`; take it first, then use the sections above to repair anything it exposes.